Building a Bookstore DB

Create Database and Tables

```
import sqlite3

# Create a database named Bookstore
conn = sqlite3.connect('Bookstore.db') cursor =
conn.cursor()

# Create Authors table
# Stores author information (AuthorID, Name) cursor.execute('''
    CREATE TABLE IF NOT EXISTS Authors (
        AuthorID INTEGER PRIMARY KEY,
        Name TEXT
    )
''')
#-----------------------------------------------------
# Create Books table
# Stores book information (BookID, Title, AuthorID, Price) cursor.execute('''
    CREATE TABLE Books (
        BookID INTEGER PRIMARY KEY AUTOINCREMENT, Title
        TEXT NOT NULL,
        AuthorID INTEGER, Price
        REAL,
        FOREIGN KEY (AuthorID) REFERENCES Authors(AuthorID)
    )
''')
#-----------------------------------------------------
# Create Sales table
# Stores sales data (SaleID, BookID, Quantity, SaleDate) cursor.execute('''
    CREATE TABLE Sales (
        SaleID INTEGER PRIMARY KEY AUTOINCREMENT,
        BookID INTEGER,
        Quantity INTEGER,
        SaleDate TEXT,
        FOREIGN KEY (BookID) REFERENCES Books(BookID)
    )
''')
```

```
<sqlite3.Cursor at 0x7a7cfd166940>
```

```
# Commit the changes conn.commit()
print("Table created successfully")
```

```
Table created successfully
```

```
# Show tables created
cursor.execute("SELECT name FROM sqlite_master WHERE type='table';") print(cursor.fetchall())
```

```
[('Authors',), ('Books',), ('sqlite_sequence',), ('Sales',)]
```

Insert Sample Data

```
    # Insert authors
    authors_data = [
        ('Mohamed',),
    ('Ali',),
    ('Ahmed',),
    ('Moaz',),
    ('Fady',)
]
cursor.executemany('INSERT INTO Authors (Name) VALUES (?)', authors_data)
```

```
<sqlite3.Cursor at 0x7a7cfd166940>
```

```
# Insert books
books_data = [
    ('Pride and Prejudice', 1, 12.99),
    ('Sense and Sensibility', 1,10.99), ('A Tale of Two
    Cities', 2, 15.99), ('Oliver Twist', 2, 14.99),
    ('The Hobbit', 3, 18.99),
    ('The Lord of the Rings', 3, 25.99),
    ('Murder on the Orient Express', 4, 11.99), ('And Then There
    Were None', 4, 9.99),
    ('The Great Gatsby', 5, 13.99),
    ('Tender is the Night', 5, 12.99)
]
cursor.executemany('INSERT INTO Books (Title, AuthorID, Price) VALUES (?, ?, ?)', books_data)
```

```
<sqlite3.Cursor at 0x7a7cfd166940>
```

```
# Insert sales data sales_data = [
    (1, 2, '2023-01-15'),
    (1, 3, '2023-02-20'),
    (2, 1, '2023-03-10'),
    (3, 5, '2023-04-05'),
    (4, 2, '2023-05-12'),
    (5, 1, '2023-06-18'),
    (6, 4, '2023-07-22'),
    (7, 2, '2023-08-01'),
    (8, 3, '2023-09-15'),
    (9, 1, '2023-10-28'),
    (10, 2, '2023-11-11'),
    (1, 1, '2023-12-05'),
    (2, 4, '2024-01-05'),
    (8, 4, '2024-01-15'),
]
cursor.executemany('INSERT INTO Sales (BookID, Quantity, SaleDate) VALUES (?, ?, ?)', sales_data)
```

```
<sqlite3.Cursor at 0x7a7cfd166940>
```

```
# Commit the changes conn.commit()

print("Data inserted successfully")
```

```
Data inserted successfully
```

```
# Retrieve all records from the 'author' table cursor.execute("SELECT * FROM
Authors ")
rows = cursor.fetchall()

print("All records in the 'author' table:") for row in rows:
    print(row)
```

```
All records in the 'author' table: (1, 'Mohamed')
(2, 'Ali')
(3, 'Ahmed')
(4, 'Moaz')
(5, 'Fady')
```

```
# Retrieve all records from the 'Books' table cursor.execute("SELECT * FROM
Books ")
rows = cursor.fetchall()

print("All records in the 'Books' table:") for row in rows:
    print(row)
    print("-"*60)
```

```
All records in the 'Books' table:
(1, 'Pride and Prejudice', 1, 12.99)
------------------------------------------------------------
(2, 'Sense and Sensibility', 1, 10.99)
------------------------------------------------------------
(3, 'A Tale of Two Cities', 2, 15.99)
------------------------------------------------------------
(4, 'Oliver Twist', 2, 14.99)
------------------------------------------------------------
```

```
(5, 'The Hobbit', 3, 18.99)
------------------------------------------------------------
(6, 'The Lord of the Rings', 3, 25.99)
------------------------------------------------------------
(7, 'Murder on the Orient Express', 4, 11.99)
------------------------------------------------------------
(8, 'And Then There Were None', 4, 9.99)
------------------------------------------------------------
(9, 'The Great Gatsby', 5, 13.99)
------------------------------------------------------------
(10, 'Tender is the Night', 5, 12.99)
------------------------------------------------------------
```

```
# Retrieve all records from the 'Sales' table cursor.execute("SELECT * FROM
Sales ")
rows = cursor.fetchall()

print("All records in the 'Sales' table:") for row in rows:
    print(row)
    print("-"*60)
```

```
All records in the 'Sales' table:
(1, 1, 2, '2023-01-15')
------------------------------------------------------------
(2, 1, 3, '2023-02-20')
------------------------------------------------------------
(3, 2, 1, '2023-03-10')
------------------------------------------------------------
(4, 3, 5, '2023-04-05')
------------------------------------------------------------
(5, 4, 2, '2023-05-12')
------------------------------------------------------------
(6, 5, 1, '2023-06-18')
------------------------------------------------------------
(7, 6, 4, '2023-07-22')
------------------------------------------------------------
(8, 7, 2, '2023-08-01')
------------------------------------------------------------
(9, 8, 3, '2023-09-15')
------------------------------------------------------------
(10, 9, 1, '2023-10-28')
------------------------------------------------------------
(11, 10, 2, '2023-11-11')
------------------------------------------------------------
(12, 1, 1, '2023-12-05')
------------------------------------------------------------
(13, 2, 4, '2024-01-05')
------------------------------------------------------------
(14, 8, 4, '2024-01-15')
------------------------------------------------------------
```

Query and Use SQL Joins

```
# Retrieve all books and their authors import pandas as pd
first_query = """ SELECT
            Books.Title,
            Authors.Name AS Author FROM
    Books
    INNER JOIN Authors ON Books.AuthorID = Authors.AuthorID
"""
result1_df = pd.read_sql_query(first_query, conn)
#-----------------------------------------------------
result1_df
```

	Title	Author
0	Pride and Prejudice	Mohamed
1	Sense and Sensibility	Mohamed
2	A Tale of Two Cities	Ali
3	Oliver Twist	Ali
4	The Hobbit	Ahmed
5	The Lord of the Rings	Ahmed
6	Murder on the Orient Express	Moaz
7	And Then There Were None	Moaz
8	The Great Gatsby	Fady
9	Tender is the Night	Fady

```
# Find total sales per book (total quantity sold) second_query = """
    SELECT
        Books.Title,
        SUM(Sales.Quantity) AS TotalQuantitySold,
        SUM(Sales.Quantity * Books.Price) AS TotalRevenue  FROM Books
    INNER JOIN Sales ON Books.BookID = Sales.BookID GROUP BY
    Books.BookID
"""
#-----------------------------------------------------
result2_df = pd.read_sql_query(second_query, conn) result2_df
```

	Title	TotalQuantitySold	TotalRevenue
0	Pride and Prejudice	6	77.94
1	Sense and Sensibility	5	54.95
2	A Tale of Two Cities	5	79.95
3	Oliver Twist	2	29.98
4	The Hobbit	1	18.99
5	The Lord of the Rings	4	103.96
6	Murder on the Orient Express	2	23.98
7	And Then There Were None	7	69.93
8	The Great Gatsby	1	13.99
9	Tender is the Night	2	25.98

```
# Combine data from Books and Sales using JOIN third_query = """
    SELECT
        Books.Title,
        Sales.Quantity,
        Sales.SaleDate,
        Books.Price
    FROM Books
    INNER JOIN Sales ON Books.BookID = Sales.BookID
"""
result3_df = pd.read_sql_query(third_query, conn)
#-----------------------------------------------------
result3_df
```

	Title	Quantity	SaleDate	Price
0	Pride and Prejudice	2	2023-01-15	12.99
1	Pride and Prejudice	3	2023-02-20	12.99
2	Sense and Sensibility	1	2023-03-10	10.99
3	A Tale of Two Cities	5	2023-04-05	15.99
4	Oliver Twist	2	2023-05-12	14.99
5	The Hobbit	1	2023-06-18	18.99
6	The Lord of the Rings	4	2023-07-22	25.99
7	Murder on the Orient Express	2	2023-08-01	11.99
8	And Then There Were None	3	2023-09-15	9.99
9	The Great Gatsby	1	2023-10-28	13.99
10	Tender is the Night	2	2023-11-11	12.99
11	Pride and Prejudice	1	2023-12-05	12.99
12	Sense and Sensibility	4	2024-01-05	10.99
13	And Then There Were None	4	2024-01-15	9.99

```
# Close the connection  conn.close()
```